Data Communication and Security (MCA40140)

CCA2 Assignment — Model Answers

MCA First Year, Semester 1 — MIT WPU | Each question carries 8 Marks
Submission: 30-10-2026, 3:00 PM

Q1. Explain the differences between OSI and TCP/IP reference models with suitable diagrams.

- **OSI model:** OSI (Open Systems Interconnection) is a 7-layer reference model developed by ISO. Its purpose is to let different systems communicate without changing their hardware or software. It is a guideline and was never fully implemented.
- **TCP/IP model:** TCP/IP is a practical 4-layer protocol suite. It became the standard of the Internet because it was widely used and tested.

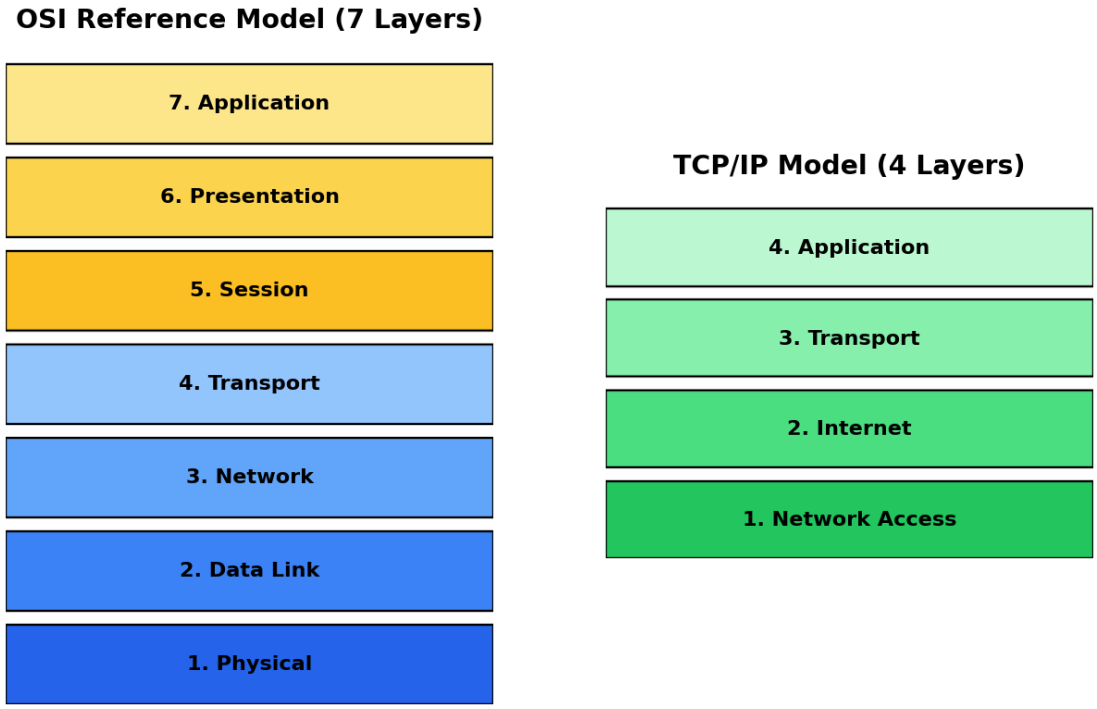


Fig: OSI model (7 layers) and TCP/IP model (4 layers)

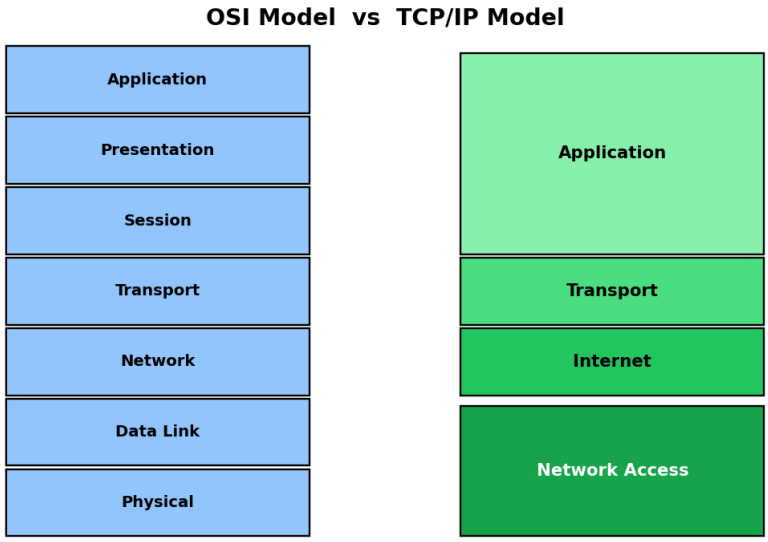


Fig: Layer-wise mapping of the OSI and TCP/IP models

Mapping: OSI Application + Presentation + Session = TCP/IP Application; OSI Transport = Transport; OSI Network = Internet; OSI Data Link + Physical = Network Access.

Parameter	OSI Model	TCP/IP Model
Full form	Open Systems Interconnection	Transmission Control Protocol / Internet Protocol
Developed by	ISO (International Organization for Standardization)	ARPANET / DARPA (US Department of Defense)
Number of layers	7 layers	4 layers
Layers	Physical, Data Link, Network, Transport, Session, Presentation, Application	Network Access, Internet, Transport, Application
Nature	Theoretical reference model (guideline)	Practical model, used in real networks
Usage	Never fully implemented	Basis of the Internet; widely implemented
Session and Presentation	Separate layers	Merged into the Application layer
Approach	Vertical, layer-by-layer; model first, protocols later	Horizontal; protocols first, model built around them
Protocol dependency	Protocol independent (generic)	Protocol dependent (TCP, IP, UDP defined)
Layer boundaries	Strict and clearly separated	Less strict, flexible
Network layer service	Supports both connectionless and connection-oriented service	Supports only connectionless service (IP)
Transport layer service	Supports only connection-oriented service	Supports both: TCP (connection-oriented) and UDP (connectionless)

Q2. Explain the switching techniques TDM and FDM with suitable example.

TDM and FDM are multiplexing (channel sharing) techniques. They allow many signals to share a single link, so that the link is used efficiently. A multiplexer (MUX) combines the signals at the sender and a de-multiplexer (DEMUX) separates them at the receiver.

1. Time Division Multiplexing (TDM):

- The time of the link is divided into small slots. Each user gets a slot in turn (round-robin). Each user uses the full bandwidth, but only during its own slot.
- One round of slots (one slot per user) is called a frame. Frames repeat continuously. TDM is mainly used for digital signals.
- **Example:** Four digital sources A, B, C, D each send 1000 bits per second (1 bit per millisecond). The MUX takes 1 bit from each source in turn, so one frame = 4 bits (A B C D) and one frame is sent every 1 ms. The link rate = 4 × 1000 = 4000 bps, and each slot lasts 0.25 ms. The DEMUX at the receiver sends slot 1 to A, slot 2 to B, and so on. Real use: digital telephone trunks.

2. Frequency Division Multiplexing (FDM):

- The total bandwidth of the link is divided into separate frequency bands. Each user is given one band and all users transmit at the same time. Each signal is modulated on a different carrier frequency.
- Unused gaps called guard bands are kept between neighbouring bands to avoid interference. FDM is mainly used for analog signals.
- **Example:** Three voice channels of 4 kHz each are multiplexed. With 1 kHz guard bands between them, the link needs 3 × 4 + 2 × 1 = 14 kHz. Channel 1 may use 20–24 kHz, channel 2 25–29 kHz and channel 3 30–34 kHz. Real use: radio and TV broadcasting, cable TV.

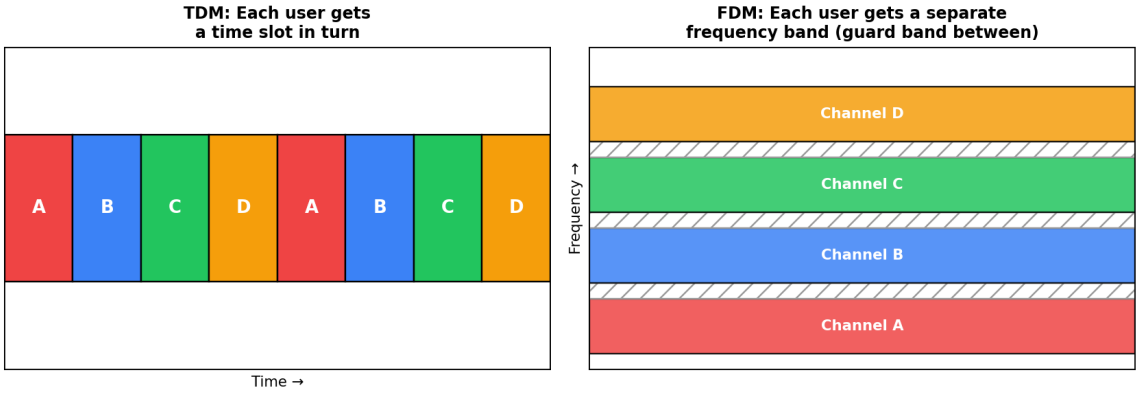


Fig: TDM (time slots) and FDM (frequency bands)

Parameter	TDM	FDM
Sharing is done by	Time (each user gets a time slot)	Frequency (each user gets a frequency band)
Transmission	Users send one after another	All users send at the same time
Bandwidth used by a user	Full bandwidth for a short time	Part of the bandwidth all the time
Guard needed	Guard time between slots	Guard band between frequency bands
Signal type	Mainly digital	Mainly analog
Circuit/Hardware	Simple, needs synchronization	Needs filters and modulators
Example	Digital telephone system	Radio, TV and cable TV

Q3. Explain the working of Cyclic Redundancy Check (CRC) with a sample binary message.

- **CRC:** Cyclic Redundancy Check is a powerful error-detection method. The sender adds extra bits (the CRC) to the data. These bits are the remainder of a binary (modulo-2) division of the data by an agreed divisor called the generator. The receiver divides the received codeword by the same generator. A zero remainder means no error was found.
- Modulo-2 arithmetic: addition and subtraction are both the same as XOR (0+0=0, 0+1=1, 1+0=1, 1+1=0). There are no carries or borrows.

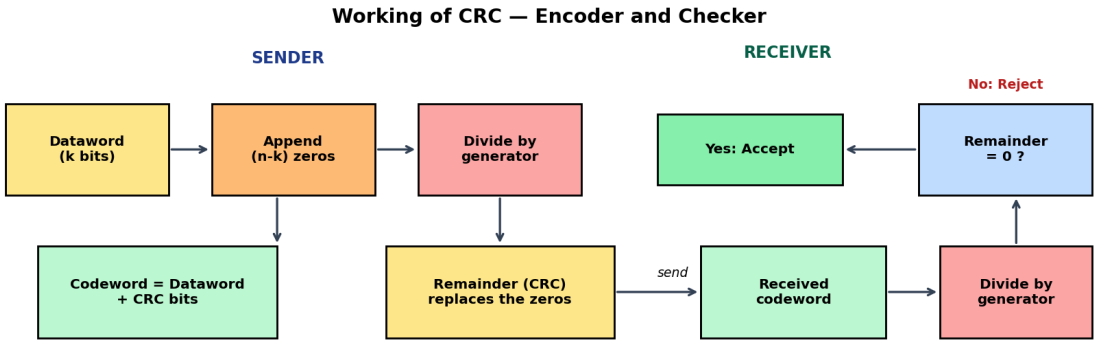


Fig: Working of CRC at the sender and receiver

Steps at the sender:

- Step 1: Choose a generator of (r + 1) bits. The CRC will be r bits long.
- Step 2: Append r zeros to the right of the dataword.
- Step 3: Divide this new number by the generator using modulo-2 division. The remainder (r bits) is the CRC.
- Step 4: Replace the appended zeros with the CRC. The result is the codeword that is transmitted.

Sample message:

Dataword = 100100 (6 bits), Generator = 1101 ($x^3 + x^2 + 1$, 4 bits), so r = 3.

Dividend after appending 3 zeros = 100100000. Divide by 1101:

Step	Bits taken	XOR with	Result	Next bit brought down
1	1001	1101	0100	0
2	1000	1101	0101	0
3	1010	1101	0111	0
4	1110	1101	0011	0
5	0110	0000	0110	0
6	1100	1101	0001	-

- **Remainder (CRC):** 001
- **Transmitted codeword:** 100100 + 001 = 100100001

At the receiver:

- **No error:** Divide 100100001 by 1101. The remainder is 000, so the data is accepted.
- **With error:** Suppose the received word is 100000001 (one bit changed). Division by 1101 gives remainder 011, which is not zero, so the data is rejected.

Advantages:

- Detects all single-bit errors, all double-bit errors (with a suitable generator), all odd numbers of errors and most burst errors.

- Easy to implement in hardware using shift registers, so it is used in Ethernet, HDLC and many other protocols.

Q4. The received Hamming codeword is 0110101010101011. Identify the position of the error using even parity checks and correct the error.

Received codeword (16 bits): 0 1 1 0 1 0 1 0 1 0 1 0 1 0 1 1

Step 1: Parity bits are at positions that are powers of 2, that is 1, 2, 4 and 8. These four parity bits cover positions 1 to 15. Each parity check covers the positions whose binary number has a 1 in that check's place.

Position	1	2	3	4	5	6	7	8
Bit	0	1	1	0	1	0	1	0

Position	9	10	11	12	13	14	15	16
Bit	1	0	1	0	1	0	1	1

Step 2: Calculate the even parity checks on the received bits. If the number of 1s in the group is odd, the check fails (value 1).

Check	Positions checked	Bits in group	Number of 1s	Result
C1	1, 3, 5, 7, 9, 11, 13, 15	0, 1, 1, 1, 1, 1, 1, 1	7 (odd)	C1 = 1
C2	2, 3, 6, 7, 10, 11, 14, 15	1, 1, 0, 1, 0, 1, 0, 1	5 (odd)	C2 = 1
C4	4, 5, 6, 7, 12, 13, 14, 15	0, 1, 0, 1, 0, 1, 0, 1	4 (even)	C4 = 0
C8	8, 9, 10, 11, 12, 13, 14, 15	0, 1, 0, 1, 0, 1, 0, 1	4 (even)	C8 = 0

Step 3: Syndrome = C8 C4 C2 C1 = 0 0 1 1 = 3. A non-zero syndrome means an error, and its decimal value is the error position.

- **Error position:** Bit 3.
- **Correction:** Bit 3 is 1, so flip it to 0.
- **Corrected codeword:** 0 1 0 0 1 0 1 0 1 0 1 0 1 0 1 1 (0100101010101011)

Verification: after correction all four checks give 0 (C1: 6 ones, C2: 4 ones, C4: 4 ones, C8: 4 ones, all even), so the syndrome is 0000.

Note: bit 16 lies outside the range covered by P1, P2, P4 and P8. It behaves as an extra overall-parity bit. The received word has 9 ones (odd), which agrees with the presence of exactly one wrong bit. After correction it has 8 ones (even).

Q5. Explain the TCP/IP Protocol Suite, detailing the functions of each layer.

TCP/IP is a hierarchical protocol suite made of interactive modules, each giving a specific function. It has four layers. Each layer uses the services of the layer below it, and data is encapsulated (a header is added) as it moves down the layers.

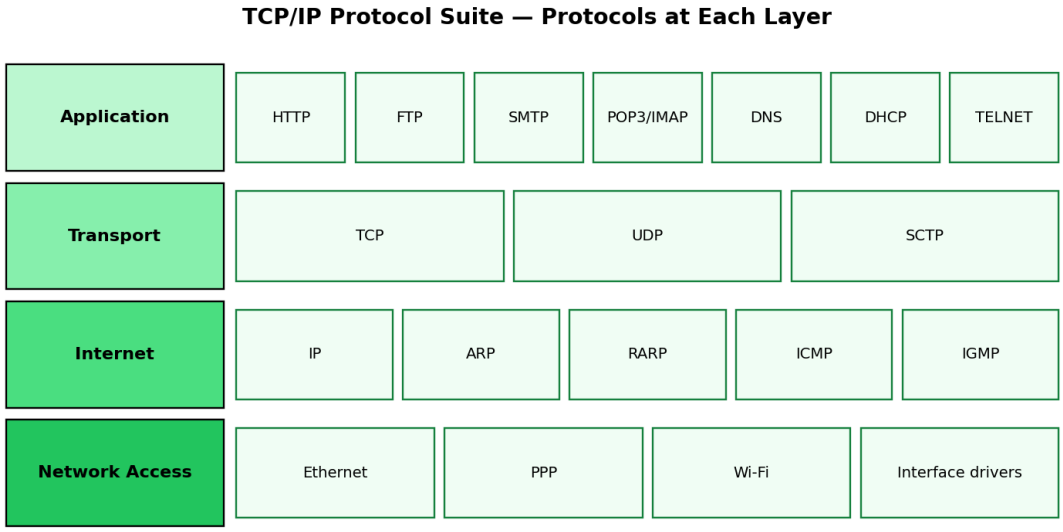


Fig: TCP/IP layers and the protocols working at each layer

Layer	Functions	Main protocols	Data unit
Application	Gives network services to the user. Combines the OSI Session, Presentation and Application layers. Handles data representation, encoding and dialog control.	HTTP, FTP, SMTP, POP3/ IMAP, DNS, DHCP, TELNET	Message
Transport	Process-to-process delivery between hosts. Divides data into segments, adds port numbers, and gives flow control and error control (in TCP).	TCP, UDP, SCTP	Segment (TCP) / Datagram (UDP)
Internet	Logical (IP) addressing and routing; finds the best path for packets across networks. IP is unreliable and connectionless (best effort).	IP, ARP, RARP, ICMP, IGMP	Packet
Network Access	Controls the hardware and the medium. Uses physical (MAC) addresses, framing and access to the physical medium. Combines OSI Physical and Data Link layers.	Ethernet, PPP, Wi-Fi	Frame / bits

Supporting protocols of the Internet layer:

- **ARP:** finds the physical (MAC) address when the IP address is known.
- **RARP:** finds the IP address when the physical address is known.
- **ICMP:** sends error and control messages about datagram problems back to the sender.

- **IGMP:** helps send a message to a group of receivers (multicast).

Parameter	TCP	UDP
Connection	Connection-oriented (connection set up first)	Connectionless
Reliability	Reliable (acknowledgments, retransmission)	Unreliable (no error correction)
Speed	Slower	Faster
Use	Web, e-mail, file transfer	Voice/video streaming, DNS

Q6. Explain HDLC protocol. Explain its different frame types and describe the importance of every field in the HDLC frame structure with a diagram.

- **Definition:** HDLC (High-Level Data Link Control) is a bit-oriented Data Link layer protocol for reliable communication over point-to-point and multipoint links. It uses bit stuffing, sequence numbers and acknowledgments (sliding window) for flow and error control.

Stations and transfer modes:

- **Primary station:** controls the link, sends commands and handles error recovery.
- **Secondary station:** works under the control of the primary and sends responses.
- **Combined station:** can act as both primary and secondary.
- **NRM (Normal Response Mode):** secondary can send only when the primary allows it (unbalanced).
- **ABM (Asynchronous Balanced Mode):** combined stations; either side can send at any time. Most widely used.
- **ARM (Asynchronous Response Mode):** secondary may start sending without permission, but the primary still controls the link. Rarely used.

HDLC frame structure:

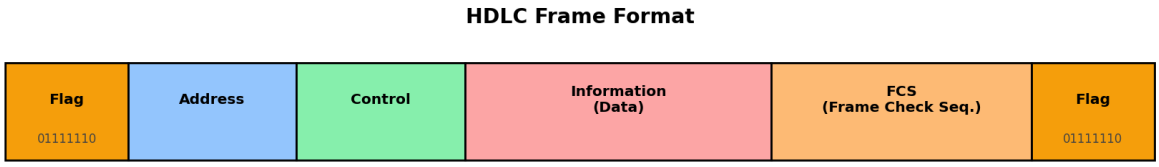


Fig: HDLC frame format

Field	Size	Importance of the field
Flag	8 bits (01111110)	Marks the start and end of the frame so the receiver can find the frame boundaries. Bit stuffing stops this pattern from appearing in the data.
Address	8 bits (can be extended)	Identifies the secondary station that is the sender (in a response) or the receiver (in a command). Needed on multipoint links.
Control	8 bits (16 in extended mode)	Shows the frame type (I, S or U) and carries sequence numbers, acknowledgment numbers and the Poll/Final bit for flow and error control.
Information	Variable length	Carries the user data (or network layer packet). Present in I-frames; absent in S-frames and in most U-frames.

Field	Size	Importance of the field
FCS (Frame Check Sequence)	16 or 32 bits	Error-detection field calculated with CRC over Address, Control and Information. A mismatch at the receiver means the frame is damaged and it is discarded.

Frame types (decided by the Control field):

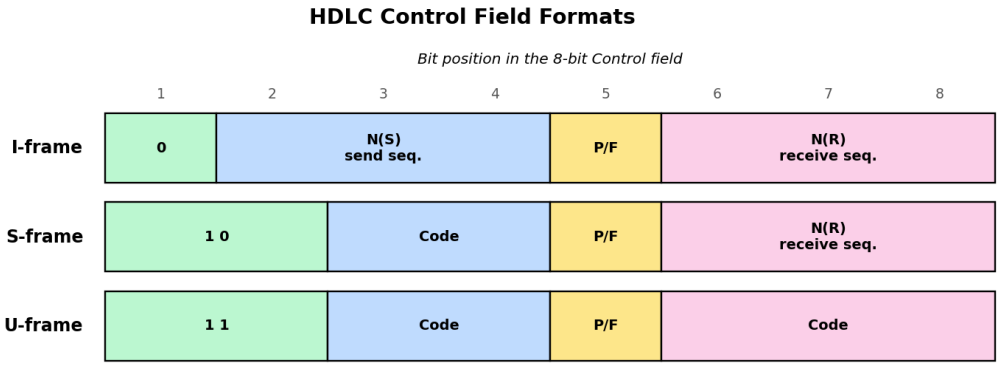


Fig: Control field formats of I-, S- and U-frames

Frame type	Purpose	Control field details
I-frame (Information)	Carries user data and also piggybacks flow/error control information.	First bit = 0. N(S) = sequence number of the frame sent; N(R) = number of the next frame expected (acts as ACK); P/F = Poll/Final bit.
S-frame (Supervisory)	Used only for flow and error control. Has no information field.	Starts with 1 0. Code gives the type: RR (Receive Ready, ACK), RNR (Receive Not Ready, stop sending), REJ (Reject, Go-Back-N), SREJ (Selective Reject). N(R) gives the ack/nak number.
U-frame (Unnumbered)	Used for link management: setting up, controlling and ending the connection. Not numbered.	Starts with 1 1. Code gives the command/response such as SNRM/SABM (set mode), DISC (disconnect), UA (unnumbered acknowledgment), FRMR (frame reject).

Q7. A data word 1001010101010 is to be transmitted using CRC with generator polynomial $X^4 + X^3 + X + 1$. Show the step-by-step division process and find out the transmitted codeword.

- **Step 1 — Generator:** $X^4 + X^3 + X + 1 = 1 \cdot X^4 + 1 \cdot X^3 + 0 \cdot X^2 + 1 \cdot X + 1$, so the generator is 11011 (5 bits). The degree is 4, so the CRC has 4 bits.
- **Step 2 — Append zeros:** Dataword = 1001010101010 (13 bits). Append 4 zeros: 10010101010100000 (17 bits). This is the dividend.
- **Step 3 — Modulo-2 division by 11011:** If the first bit is 1, XOR with 11011; if it is 0, XOR with 00000. Then drop the first bit and bring down the next bit.

Step	Bits taken	XOR with	Result	Next bit brought down
1	10010	11011	01001	1
2	10011	11011	01000	0
3	10000	11011	01011	1
4	10111	11011	01100	0

Step	Bits taken	XOR with	Result	Next bit brought down
5	11000	11011	00011	1
6	00111	00000	00111	0
7	01110	00000	01110	1
8	11101	11011	00110	0
9	01100	00000	01100	0
10	11000	11011	00011	0
11	00110	00000	00110	0
12	01100	00000	01100	0
13	11000	11011	00011	-

- **Step 4 — Remainder:** After the last step the remainder (CRC) = 0011
- **Step 5 — Codeword:** Replace the 4 appended zeros by the CRC: 1001010101010 + 0011 = 10010101010100011

Transmitted codeword = 1 0 0 1 0 1 0 1 0 1 0 1 0 0 0 1 1 (10010101010100011)

Check at the receiver: dividing 10010101010100011 by 11011 gives remainder 0000, so the codeword is accepted.

Q8 (a). Explain multiplexing in data communication. Explain the working principle of Time Division Multiplexing (TDM), differentiate between Synchronous TDM and Asynchronous TDM, with proper diagram.

- **Multiplexing:** Multiplexing is the process of combining multiple signals into one signal over a shared medium. It divides one communication channel into several logical channels, each used for a different message signal or data stream.
- The device that combines the signals is the Multiplexer (MUX). The reverse process at the receiver, extracting the individual signals from the one channel, is called De-multiplexing and is done by the De-multiplexer (DEMUX).
- Need: one expensive high-capacity link is shared by many users, so cost falls and bandwidth is used efficiently. Types: Time Division Multiplexing (TDM) and Frequency Division Multiplexing (FDM).

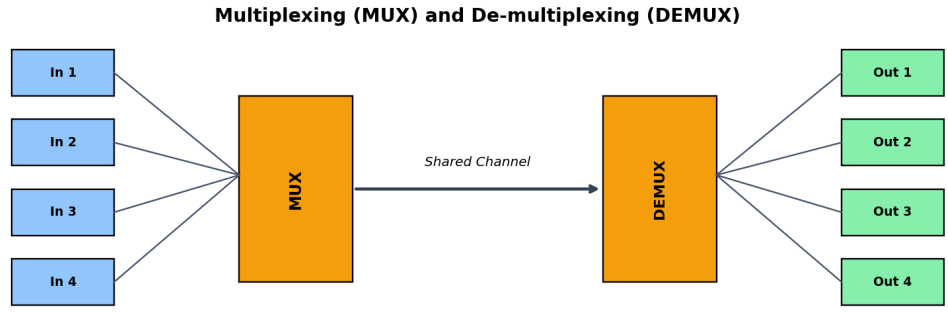


Fig: Multiplexer (MUX) and De-multiplexer (DEMUX)

Working principle of TDM:

- TDM is a multiplexing technique in which each signal is assigned a different time slot for transmission. Users share the total available bandwidth on a time-sharing basis.
- The time domain is divided into recurring slots of fixed length. The MUX visits the inputs one by one (round-robin), takes a small unit of data (bit or byte) from each and places it in that input's slot.

- One complete round of slots, one slot per input, forms a frame. Frames are sent one after another on the link.
- At the receiver, the DEMUX separates the slots of every frame and delivers each slot to the correct output line. Each user gets the full bandwidth, but only for its own time slot.
- Guard time and synchronization bits are added so the receiver can find the start of each frame and slot correctly.

Types of TDM:

- **Synchronous TDM:** Time slots are pre-assigned and fixed. A slot is given to a source even if it has no data at that time, and in that case the slot is transmitted empty. No address is needed, because the position of the slot identifies the source. Used for multiplexing digitized voice streams.

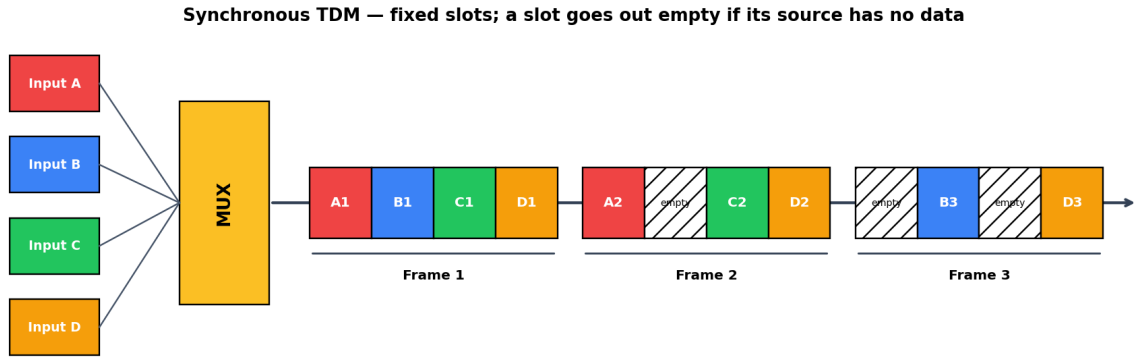


Fig: Synchronous TDM block diagram (empty slots are sent when a source is idle)

- **Asynchronous (Statistical) TDM:** Slots are allocated dynamically, depending on the speed and ready state of each source. Only sources that have data get a slot, so no empty slots are sent and channel capacity is saved. Since position no longer identifies the source, every slot carries an address.

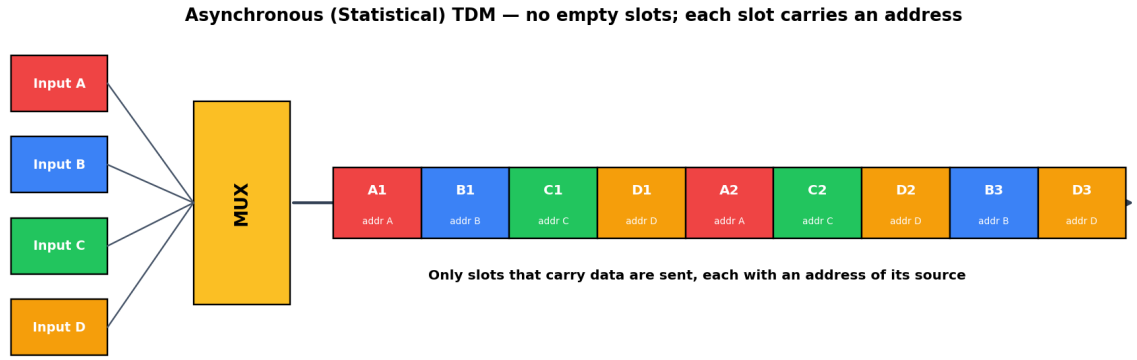


Fig: Asynchronous (Statistical) TDM block diagram

Difference between Synchronous and Asynchronous TDM:

Parameter	Synchronous TDM	Asynchronous (Statistical) TDM
Slot allocation	Fixed, pre-assigned to each input	Dynamic, given only to inputs that have data
Empty slots	Sent empty if the source is idle	Not sent; no empty slots
Addressing	Not needed (slot position identifies source)	Needed (each slot carries source address)
Bandwidth utilization	Poor when sources are idle (capacity wasted)	Efficient, idle time is used by other sources

Parameter	Synchronous TDM	Asynchronous (Statistical) TDM
Link data rate	Equal to the sum of all input rates	Can be less than the sum of input rates
Complexity and overhead	Simple, low overhead	More complex; needs buffers, addressing overhead
Typical use	Digitized voice, telephone trunks	Data traffic and bursty sources

Q8 (b). Compare static and dynamic channel allocation techniques, and explain the working principles, collision management, and throughput performance of Pure ALOHA and Slotted ALOHA.

- **Channel allocation problem:** When many stations share a single broadcast channel (as in LANs and wireless networks), the MAC sublayer must decide who may transmit and when. If two stations transmit at the same time, their frames collide and are lost. The problem is how to divide one channel among competing users fairly and efficiently.

Parameter	Static (Fixed) Allocation	Dynamic Allocation
Method	Channel is divided permanently into N parts (FDM or TDM), one for each user	Channel is given to users on demand, using access protocols
Collisions	None, since each user has its own part	Possible, so collision handling is needed
Efficiency	Poor for bursty traffic; idle users waste their share	High; idle capacity is used by busy users
Complexity	Simple	More complex (access protocol needed)
Examples	Telephone trunks using FDM/TDM	ALOHA, CSMA, token passing, Wi-Fi

Pure ALOHA:

- Working: any station sends a frame whenever it has data, without checking the channel. After sending, it waits for an acknowledgment.
- Collision management: if two frames overlap even by one bit, both are destroyed and no ACK comes. After a time-out the station waits for a random back-off time and retransmits. The random wait avoids the same collision again. After a maximum number of attempts the station gives up.
- Vulnerable time = 2 × Tt (Tt = frame transmission time), because a frame can be hit by one that started up to Tt before or up to Tt after it.

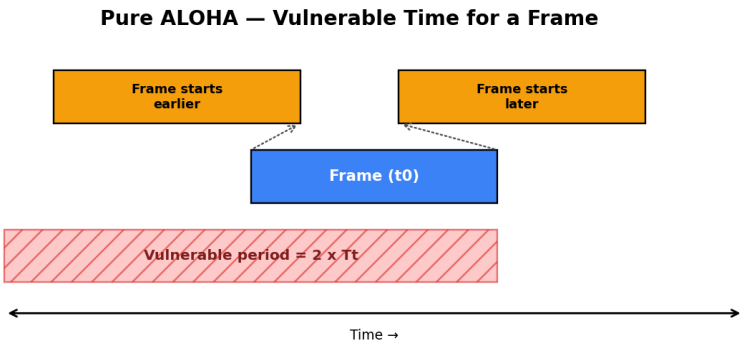


Fig: Vulnerable period of Pure ALOHA

- Throughput: S = G × e^(−2G), where G = average number of frames generated per frame time. This is G multiplied by the probability that no other frame is sent in the vulnerable time (e^(−2G)). Maximum S = 1/(2e) ≈ 0.184 (18.4 %) at G = 0.5.

Slotted ALOHA:

- Working: time is divided into slots equal to one frame time. All stations are synchronized, and a frame can be sent only at the beginning of a slot.
- Collision management: frames either overlap completely or do not overlap at all, so a collision wastes only one slot. A station that collides waits a random number of slots and tries again.
- Vulnerable time = Tt (half of Pure ALOHA).
- Throughput: S = G × e^(−G). Maximum S = 1/e ≈ 0.368 (36.8 %) at G = 1, which is double that of Pure ALOHA.

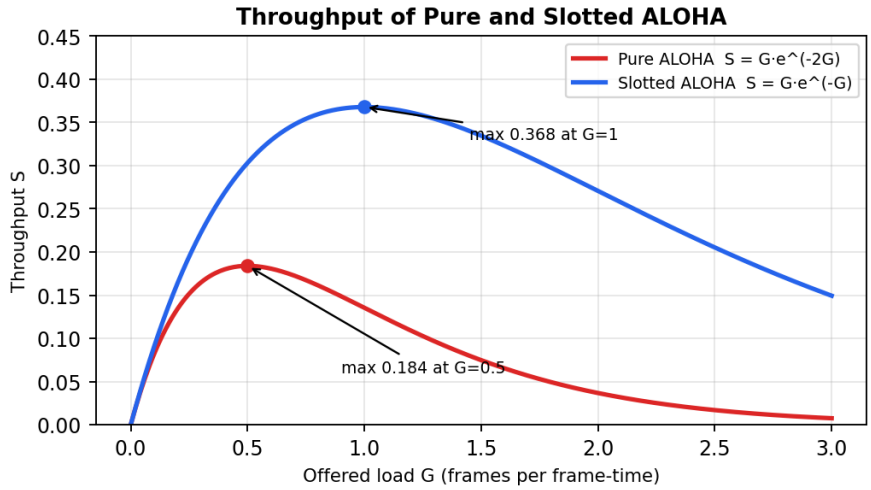


Fig: Throughput versus offered load for Pure and Slotted ALOHA

Parameter	Pure ALOHA	Slotted ALOHA
Transmission time	Any time	Only at the start of a slot
Synchronization	Not needed	Needed (common clock)
Vulnerable time	2 × Tt	Tt
Throughput formula	S = G·e^(−2G)	S = G·e^(−G)
Maximum throughput	18.4 % (at G = 0.5)	36.8 % (at G = 1)
Complexity	Simple	Slightly complex

Q9. An 11-bit Hamming codeword 10101011011 has been received. Using the Hamming method, detect and correct any error present. Show all calculation steps clearly.

Received codeword (11 bits): 1 0 1 0 1 0 1 1 0 1 1

Step 1: Parity bits are at positions 1, 2, 4 and 8 (4 parity bits + 7 data bits). Data bits are at positions 3, 5, 6, 7, 9, 10, 11.

Position	1	2	3	4	5	6	7	8	9	10	11
Bit	1	0	1	0	1	0	1	1	0	1	1

Step 2: Even parity checks. If the number of 1s in a group is odd, the check fails (value 1).

Check	Positions checked	Bits in group	Number of 1s	Result
C1	1, 3, 5, 7, 9, 11	1, 1, 1, 1, 0, 1	5 (odd)	C1 = 1
C2	2, 3, 6, 7, 10, 11	0, 1, 0, 1, 1, 1	4 (even)	C2 = 0
C4	4, 5, 6, 7	0, 1, 0, 1	2 (even)	C4 = 0
C8	8, 9, 10, 11	1, 0, 1, 1	3 (odd)	C8 = 1

Step 3: Syndrome = C8 C4 C2 C1 = 1 0 0 1 = 9 (decimal).

- **Error position:** Bit 9 (not zero, so an error is present).
- **Correction:** Bit 9 is 0, so flip it to 1.
- **Corrected codeword:** 1 0 1 0 1 0 1 1 1 1 1 (10101011111)

Verification: C1 = 1,1,1,1,1,1 (six 1s, even), C2 = 0,1,0,1,1,1 (four 1s, even), C4 = 0,1,0,1 (two 1s, even), C8 = 1,1,1,1 (four 1s, even). All checks are 0, so the syndrome is 0000 and the codeword is correct.

Original data bits (positions 3, 5, 6, 7, 9, 10, 11) = 1 1 0 1 1 1 1.

Q10. A user wants to send data bits 1011110101101. Find out the Hamming codeword using even parity bits and odd parity bits separately.

Data bits (m) = 13: 1 0 1 1 1 1 0 1 0 1 1 0 1

Number of parity bits r: $2^r \geq m + r + 1$. For r = 4: $16 \geq 18$ is false. For r = 5: $32 \geq 19$ is true. So r = 5 and the codeword length n = 13 + 5 = 18 bits.

Parity bits P1, P2, P4, P8, P16 go to positions 1, 2, 4, 8, 16. The 13 data bits fill positions 3, 5, 6, 7, 9, 10, 11, 12, 13, 14, 15, 17, 18 in order.

Position	1	2	3	4	5	6	7	8	9
Bit	P1	P2	1	P4	0	1	1	P8	1

Position	10	11	12	13	14	15	16	17	18
Bit	1	0	1	0	1	1	P16	0	1

Each parity bit checks the positions that have a 1 in its own binary place (P1: positions 1,3,5,7,...; P2: 2,3,6,7,10,11,...; P4: 4–7, 12–15; P8: 8–15; P16: 16–18). Count the 1s in the data positions of each group:

Parity bit	Data positions in the group	Bits	Number of 1s	Even parity bit	Odd parity bit
P1	3,5,7,9,11,13,15,17	1,0,1,1,0,0,1,0	4	0	1
P2	3,6,7,10,11,14,15,18	1,1,1,1,0,1,1,1	7	1	0
P4	5,6,7,12,13,14,15	0,1,1,1,0,1,1	5	1	0
P8	9,10,11,12,13,14,15	1,1,0,1,0,1,1	5	1	0
P16	17,18	0,1	1	1	0

Rule: even parity — the parity bit makes the total number of 1s in the group even (parity bit = 1 if the count is odd). Odd parity — the parity bit makes the total odd (parity bit = 1 if the count is even).

(a) Codeword using EVEN parity:

Position	1	2	3	4	5	6	7	8	9
Bit	0	1	1	1	0	1	1	1	1

Position	10	11	12	13	14	15	16	17	18
Bit	1	0	1	0	1	1	1	0	1

- **Even-parity Hamming codeword:** 0 1 1 1 0 1 1 1 1 1 0 1 0 1 1 1 0 1 (011101111101011101)

(b) Codeword using ODD parity:

Position	1	2	3	4	5	6	7	8	9
Bit	1	0	1	0	0	1	1	0	1

Position	10	11	12	13	14	15	16	17	18
Bit	1	0	1	0	1	1	0	0	1

- **Odd-parity Hamming codeword:** 1 0 1 0 0 1 1 0 1 1 0 1 0 1 1 0 0 1 (101001101101011001)

In both codewords the data bits stay the same at positions 3, 5, 6, 7, 9, 10, 11, 12, 13, 14, 15, 17, 18. Only the five parity bits are different.

Q11. Explain the Sliding Window Protocol and the Go-Back-N ARQ mechanism, illustrating the concepts with clear and properly labelled diagrams.

A. Sliding Window Protocol:

- **Concept:** Instead of waiting for an ACK after every frame (as in Stop-and-Wait), the sender is allowed to send several frames, up to a window size, before it must stop and wait. This is called pipelining and it keeps the link busy.
- Frames are numbered with sequence numbers (m bits, so numbers 0 to 2^m − 1, then repeat).
- Sender window: the range of sequence numbers the sender may send. It contains frames that are sent but not yet acknowledged, and frames that can still be sent.
- When an ACK arrives, the window slides forward (moves to the right) and new frames can be sent.
- Receiver window: the range of frames the receiver is willing to accept; it also slides forward as frames are received correctly.

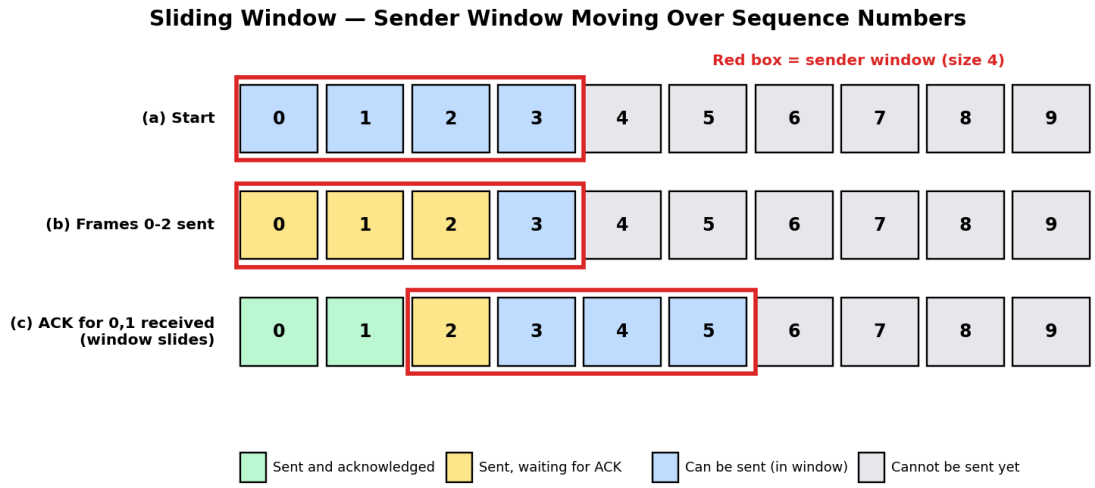


Fig: The sender window slides forward as acknowledgments arrive

Parameter	Stop-and-Wait	Sliding Window
Frames in transit	Only one	Many (up to the window size)
Link use	Poor, sender is idle while waiting	Good, link kept busy
Throughput	Low	High
Implementations	Stop-and-Wait ARQ	Go-Back-N ARQ, Selective Repeat ARQ

B. Go-Back-N ARQ:

- **Concept:** Go-Back-N is a sliding-window ARQ (Automatic Repeat reQuest) method. The sender can send up to N frames without waiting for ACKs. The receiver accepts frames only in the correct order.
- **Window sizes:** Sender window size = 2^m − 1 (for m = 3 bits, the size is at most 7). Receiver window size = 1. The sender window cannot be 2^m, because then the receiver could not tell a new frame 0 from a re-sent old frame 0.

Working:

- The sender sends frames 0, 1, 2, ... and keeps a copy of every frame that is not yet acknowledged. A timer runs for the oldest unacknowledged frame.
- The receiver checks each frame. If it is the expected frame, it is accepted and an ACK is sent (the ACK number names the next frame expected, so one ACK can confirm many frames: cumulative ACK).
- If a frame is lost or damaged, the receiver discards it and also discards all later frames (they are out of order), and sends no new ACK.
- When the timer expires, the sender goes back and resends the lost frame and all frames sent after it, even though some of them arrived correctly. This gives the name Go-Back-N.

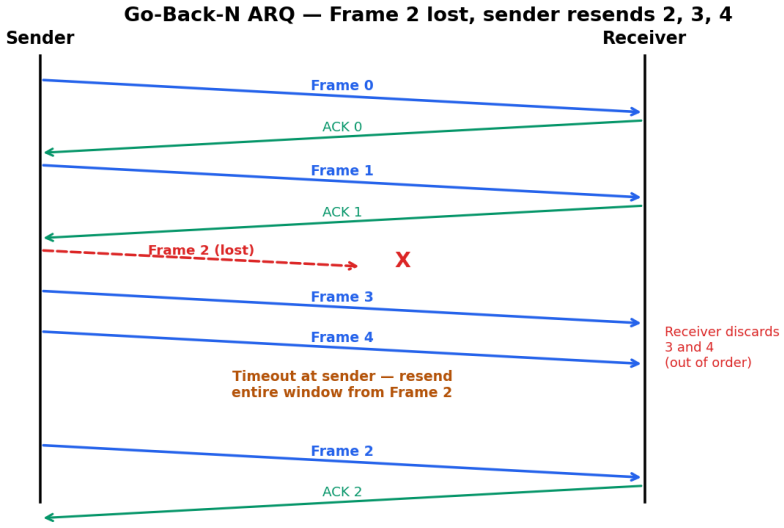


Fig: Go-Back-N ARQ — Frame 2 is lost, so Frames 2, 3 and 4 are sent again

- **Example:** Frames 0 to 4 are sent. Frame 2 is lost. Frames 3 and 4 arrive but are discarded. After the time-out the sender resends 2, 3 and 4.

Advantages and disadvantages:

- Advantage: much better link use than Stop-and-Wait; simple receiver (no buffering of out-of-order frames); cumulative ACKs reduce the number of ACKs.
- Disadvantage: wastes bandwidth on noisy links, since good frames after a lost frame are sent again.

Q12. Explain guided and unguided transmission media in data communication.

Transmission media is the communication channel that carries information from the sender to the receiver. It works at the Physical layer. Media are of two types: guided (wired) and unguided (wireless).

A. Guided (wired) media:

The signal travels through a physical cable that guides it from one device to the other.

- **1. Twisted Pair Cable:** Two insulated copper wires twisted together to reduce crosstalk and noise. UTP (Unshielded) is cheap, easy to install, has a range below 100 m and is used in Ethernet (10BaseT). STP (Shielded) has a metal-mesh shield, so it has less noise and a slightly longer range. Disadvantage: high attenuation and limited distance. Use: telephone lines, LANs.
- **2. Coaxial Cable:** A central copper conductor, insulation, a metal mesh shield and an outer jacket. Range about 1 km. Less noise and crosstalk and higher bandwidth than twisted pair, but costlier and bulkier. Use: cable TV, broadband.
- **3. Fiber Optic Cable:** A glass or plastic core with cladding and a jacket. It carries data as light pulses (light pulse = 1, no pulse = 0) using reflection. Very high bandwidth, very low noise, long distance, secure. Disadvantage: costly, fragile, skilled installation. Use: high-speed backbones and long-distance links.

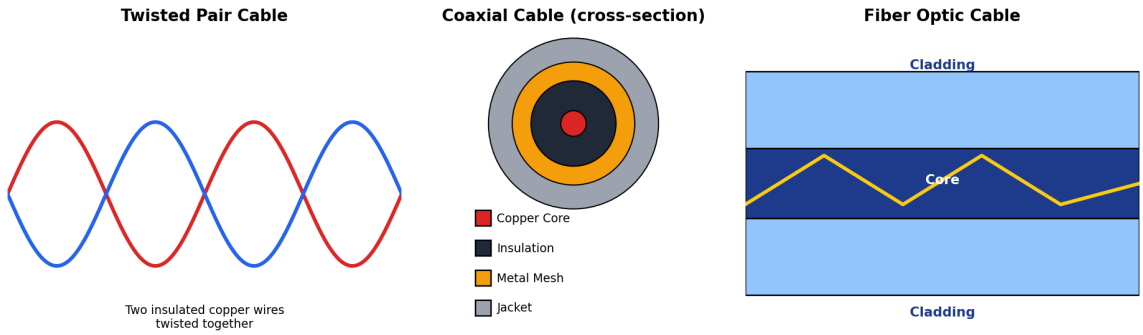


Fig: Twisted pair, coaxial and fiber optic cable

B. Unguided (wireless) media:

The signal travels through free space as electromagnetic waves, without a physical conductor. Signals are broadcast and can be received by anyone in range, so it is less secure.

Type	Frequency range	Direction	Use
Radio waves	3 kHz – 1 GHz	Omnidirectional (antennas need not be aligned); higher data rate	Radio and TV broadcasting
Microwaves	1 GHz – 300 GHz	Unidirectional (antennas must be aligned); distance depends on antenna height	Cellular phones, satellite links, wireless LANs
Infrared	300 GHz – 400 THz	Short range, line-of-sight; cannot pass through walls; high security	TV remote, wireless mouse

Difference between guided and unguided media:

Parameter	Guided Media	Unguided Media
Definition	Uses a physical medium (cable) to carry the signal	Uses free space (air); no physical medium
Also called	Wired media	Wireless media

Parameter	Guided Media	Unguided Media
Signal path	Confined along the cable	Spreads in all directions or as a beam
Examples	Twisted pair, coaxial, fiber optic	Radio waves, microwave, infrared, satellite
Security	More secure	Less secure; signal can be intercepted
Installation cost	Higher; cabling needed	Lower; no cabling, supports mobility
Interference	Less affected (especially fiber)	More affected by weather and obstacles
Distance	Limited by cable length and attenuation	Can cover very large distances (satellite)